SMART PARKING SYSTEM USING IoT AND MACHINE LEARNING

Phase 1 – Project Proposal

Course	ML 402331 – IoT Machine Learning Project
Track	Hardware Track
Student	Mohammed Alkasabreh
Team	Individual project (instructor permission obtained)

1. Project Overview

This project proposes a small-scale Smart Parking System combining IoT hardware and Machine Learning. An ESP32 will read two infrared (IR) sensors installed sequentially at the parking entrance. The activation order will be used to detect vehicle entry and exit and update the estimated occupancy of a parking area with a capacity of four vehicles. An SG90 servo will represent the entrance barrier, and an I2C LCD will display occupancy and available spaces. Sensor events and timestamps will be recorded for analysis.

2. Problem Statement

Drivers in busy areas may spend unnecessary time searching for available parking spaces. This can contribute to traffic congestion, wasted fuel, and a poor parking experience. A low-cost system that detects vehicle movement and displays parking availability may help users know whether the parking area has remaining capacity.

3. Research Question

Can an IoT-based parking system using an ESP32 and sequential IR sensors collect vehicle movement data and support classification of parking availability as Available or Full?

4. Proposed Hardware and Operation

• ESP32 development board as the controller.

• Two IR sensors placed sequentially at the entrance to detect direction from activation order.

• SG90 servo motor to demonstrate barrier movement.

• I2C LCD to display occupied and available spaces.

• Parking capacity: four vehicles; occupancy counter constrained to 0–4.

Limitation: The two entrance sensors estimate total occupancy by counting entry and exit events; they do not detect the status of each individual parking bay. The system will use debounce and sequence validation to reduce duplicate or invalid counts.

5. Machine Learning Task

Task: Binary Classification. **Target:** Parking_Status. **Classes:** Available (0), when occupancy is 0–3; Full (1), when occupancy is 4.

Candidate features include IR sensor states, detected event direction, time interval between sensor activations, and entry/exit event counts within a defined time window. Features will be reviewed to avoid target leakage. A simple occupancy rule will be retained as a baseline, and the report will discuss whether ML adds value beyond that rule.

6. Data Collection Plan

The ESP32 will record sensor events with timestamps during controlled vehicle-entry and exit tests. Each passage will be checked against the known vehicle movement and occupancy state. Records will be stored in CSV format and labeled using verified occupancy: Available for occupancy 0–3 and Full for occupancy 4.

• Target: at least 500 labeled records.

• Collection: repeated controlled entry/exit trials at different occupancy levels and under repeated sensor activation conditions.

• Fields: timestamp, IR1 state, IR2 state, detected event (entry/exit/none), occupancy count, available-space count, and Parking_Status.

• Quality checks: debounce, direction validation, occupancy limits 0–4, and checks for missing or inconsistent records.

The dataset will be split into training and testing subsets before evaluation. Classification will be assessed using accuracy, precision, recall, F1-score, and a confusion matrix. Actual collection conditions and limitations will be documented.

7. Tools and Software

• Arduino IDE for ESP32 firmware.

• Python and Google Colab for data processing and model training.

• Pandas, NumPy, Matplotlib, and scikit-learn.

• GitHub for version control and documentation.

8. Bill of Materials (Estimated)

Component	Qty	Estimated cost (JOD)
ESP32 board	1	6–10
IR sensors	2	2–4 total
SG90 servo	1	2–3
I2C LCD	1	3–5
Breadboard	1	1–2
Jumper wires	As needed	1–2
USB cable / power supply	1	1–3
Estimated total		16–29

Costs are preliminary estimates and should be updated using actual local prices. The servo may need a separate regulated 5V supply with common ground. Check IR sensor output voltage before connecting it to ESP32 GPIO pins.

9. Related Works

The following works are relevant background for smart parking and IoT system design. Verify the full bibliographic details against the publisher or library record before final submission.

1. Idris, M. Y. I., et al. (2009). “Car Park System: A Review of Smart Parking System and Its Technology.” *Information Technology Journal, 8*(2), 101–113.

2. Geng, Y., & Cassandras, C. G. (2013). “New Smart Parking System Based on Resource Allocation and Reservations.” *IEEE Transactions on Intelligent Transportation Systems, 14*(3), 1129–1139.

3. Khanna, A., & Anand, R. (2016). “IoT Based Smart Parking System.” In *2016 International Conference on Internet of Things and Applications (IOTA)*.

10. Expected Outcomes

The expected prototype will detect vehicle entry and exit, maintain an estimated occupancy count, display parking availability, and demonstrate a servo-controlled barrier. It will also produce a labeled sensor-event dataset and a comparison of classification models. The final evaluation will discuss sensor reliability, data quality, model performance, and limitations of estimating occupancy from entrance events.

11. Individual Work Plan

The project will be completed individually with instructor permission. The student will handle hardware assembly, ESP32 programming, controlled data collection, dataset preparation, machine learning experiments, evaluation, documentation,

and final demonstration.

12. References

Idris, M. Y. I., Leng, Y. Y., Tamil, E. M., Noor, N. M., & Razak, Z. (2009). Car park system: A review of smart parking system and its technology. *Information Technology Journal, 8*(2), 101–113.

Geng, Y., & Cassandras, C. G. (2013). New smart parking system based on resource allocation and reservations. *IEEE Transactions on Intelligent Transportation Systems, 14*(3), 1129–1139.

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Submission note: This is a Phase 1 draft. Replace estimated costs with actual prices where possible, verify reference metadata, and update the data plan after hardware testing.